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Autonomous robot for cleaning photovoltaic panels in desert zones ☆

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ABSTRACT

In desert zones, a continuous cleaning activity of photovoltaic panels in solar plants is required since the deposition of both airborne dust and sand after a storm can reduce their efficiency up to 80%. Manual cleaning of the photovoltaic panels in dry areas is costly, cannot make use of water and workers must be employed several times in a month, often under extreme environmental conditions. For all these reasons, the research of cleaning solutions performed by autonomous robotic systems are seen beneficial to recover the solar panels efficiency at reasonable costs also nightly. In this respect, this paper presents the implementation of an unmanned low-cost robotic device operating without rails or guides for waterless dust and sand removal from the surface of photovoltaic panels. The robot autonomously moves as a half-track and gently wipes the dust away by means of a couple of independent helical brushes that alternatively rotate, according to a defined combination of the cleaning and motion strategy. Moreover, in real-time the robot detects its spatial position by ultrasonic sensors, regulates both its speed rate and motion direction. The on-board control system, based on an ARDUINO DUE platform, settles the robot travel direction, its speed and which brush must be activated for cleaning. Experimental tests have been performed by fabricating a prototype of the device to assess its effectiveness and reliability in terms of the autonomous motion, dust removal and low power consumption.

1. Introduction

Recent studies reported improvements of the Photovoltaic Panels (PVP) efficiency by the implementation of new materials [1], processes [2] and electronic control techniques [3]. Due to the large amount of the solar energy to be converted in electrical power, the PVP efficiency (i.e., the ratio between the electrical output power and the incident solar radiation flux per unit collector area) is continuously measured by several sensors [4–8]. Nevertheless, the value of the PVP efficiency measured in laboratory standard test conditions often deviates from that one obtained in real outdoor environments [9]. Under similar operating conditions, in fact, environmental aspects such as the air temperature, the presence and speed of wind, the season of the year as well as the latitude and longitude of the solar plant location cause the lowering of the actual efficiency since all these parameters affect the incident solar radiation flux. Apart from these environmental aspects that, in general, cannot be controlled, it has been demonstrated that the major source of reduction of the PVP efficiency is the deposition of different kinds of particles on their surface that can strongly decrease the amount of the solar radiation impinging on the PVP. These particles differ in size and in their chemical/physical characteristics. Depending on the particular solar plant location, they may include soil, sand, bird droppings, carbon particles, clay and air pollutants [10–12]. As mentioned before, even if no particular solutions can be adopted to counteract the environmental factors, several solutions based on different techniques are available to clean the PVP surface. A first class of cleaning techniques makes use of water: rainfall [13] is the most obvious one but it is unpredictable especially during summer or completely absent in dry and desert regions. For this, the cleaning of the PVP using water cannot be regularly scheduled, it is not efficient after long dry periods when dust encrustations are produced or if dust is composed by insoluble particles. However, if water is available [13], the cleaning procedure can use water jets apparatus for dust scrubbing or guided moving systems equipped with brushes [14] without or with a suction system to adhere on the PVP surface [15]. Other cleaning techniques replace water with compressed air for wiping away also dust of insoluble particles [13,14]. Furthermore, all these solutions require manual labour that can be costly and applicable only in solar plants with a limited extension. Finally, some cleaning solutions have been proposed [14] that do not need manual work and external removing agent like water [13]. Dust fall out can be obtained by mounting the PVP on controllable rotating frames providing the PVP overturning. This particular cleaning solution requires

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high investment and maintenance costs in the case of large-size solar plants. On the other hand, chemical hydrophobic self-cleaning coatings can be deposited on the PVP surface, but this solution is totally inefficient in absence of water. All the above-mentioned cleaning methods cannot be applied in solar farms located in desert zones where sand accumulation on the PVP surfaces are mainly due to sandstorms that can cause an electric power efficiency reduction up to 80% [11,14]. For this reason, these large-size solar farms need continuous PVP specific cleaning activities for dust removal avoiding the use of dedicated personnel and/or the installation of specific equipment for the PVP motion or of compressed air distribution systems, both requiring high investment, operating and maintaining costs. As in the case to fulfil some demanding activities in agriculture [16–18], robotic and, in general, automatic systems are the ideal solutions also for PVP cleaning in desert zones: they can operate during evening or night when the solar plants are not operating with a reduced human intervention and maintenance problems.

As proposed in [19], three are the categories of automatic systems, all of them adopting brushes for sand removal. To the first category belong systems that need one or more guide rails permanently fixed on the PVP array. In particular, the NO water Mechanical Automated Dusting Device (NOMADD) [20], designed to operate in the desert harshest conditions, is constituted of a cleaning brush covering the height of two PVP rows and is moved by a couple of two direct drive motors: one of these provides for the brush rotation while the other one allows for the brush support frame translation achieved by employing a couple of guide rails. A similar device has been proposed in [13]. On the other hand, Ecoppia E4 [21] is a robot with two degrees of freedom: a main frame moves by using a guide rail along an array made of two PVP rows. A second frame, equipped with two cleaning and rotating brushes, moves up and down along each PVP array column. A special microfiber gently wipes soiling away downwards and off the panel rows. The second category of automatic systems contains tethered robot systems that require two or more cables to connect and move the automatic cleaning system on the PVP array [22]. However, these are not very robust systems due to the use of cables that need, in turn, additional poles to be held. The third category is specific of those mobile systems that do not require any additional equipment to move and clean the PVP arrays. Ecoppia T4 [21] is a light-weight robot that uses a water-free cleaning method combining a rotation of soft microfiber elements and the generation of a controlled airflow to push dust particles off the solar panels. This system uses a series of sensors and integrated technologies, detects the structure edges and navigates its way on the tracker using the optimal, most efficient and accurate pre-defined path. The robot is capable of traveling across the tracker central gap and on to the neighboring table using simple connecting bridges. Miraikikai Inc. [23] developed the Solar Cleaning Robot, an autonomous device moved by three wheels and equipped with two simultaneously working brushes to wipe away the sand deposited on the PVP surface. An electronic system acquires the signals generated by four sensors to control the robot motion performed with the aid of a guidance system made of mechanical pins set along the guides located between adjacent PVPs. This guidance system fixes the robot main body on the PVP structure and ensures the robot travels along a straight line and the 90° turn for its relocation on the adjacent uncleaned PVP array row/column. An inclination sensor enables or disables the cleaning activity to avoid slipping and falling. Raybot [24], by Ecovac Robotics, is a self-guided robot equipped with brushes, blowers and a dustbin to collect the dust. A set of suction cups is used to prevent damages of the solar panel and to avoid slipping and falling up to a tilting angle of 75°. Moreover, intelligent sensors are employed for the PVP edge detection. Finally, some researchers [19] proposed a cleaning robot using four omni wheels installed with a 45° angle respect to the heading axle of the robot. The robot is equipped with a motor for each wheel and with a series of encoders to control the robot motion. Photosensors are adopted to prevent the robot to fall off the PVP array. Four brushes, placed in between the wheels, provide for the sand removal.

To pursue the design and realization of a self-guided mechanized device, the fulfillment of some targets must be considered [13]: (i) the reduction of the capital and maintenance costs; (ii) the possibility of a high cleaning efficiency even of uneven PVP surface; (iii) the minimization of the energy consumption by using very efficient motors for both the robot and brush movements.

The robot proposed in this paper belongs to the third category of mobile systems. In the following we present the design, the construction and the characterization of a self-guided robot prototype. It moves by a couple of high friction rubber belts that act as two tracks powered by two independent motors. The waterless sand removal is carried out by a couple of independently motorized helical brushes placed one in the front and the other in the rear part of the robot. The implemented motion strategy implies the robot travels horizontally forward and backwards along the rows of the PVP array, starting from the higher row and ending to the lower one. Only the brush facing the motion direction wipes away forward and downward the sand until it falls down from the PVP array. The robot is equipped with ultrasonic sensors to detect and regulate in real-time its speed rate as well as its position with respect to a set of vertical barriers placed around the PVP array. A low-cost ARDUINO DUE platform [7] is at the basis of the electronic on-board control system.

In terms of the technical aspect, respect to the state-of-the art, the proposed robot introduces the following novelties: (i) it does not need any guide rails or pins or suction cups to be attached and move on the PVP array. Thus, the robot is totally autonomous also on uneven surfaces or in solar farms positioned on wavy ground where the installation and use of guides are difficult to be implemented or upgraded; (ii) unlike the other mobile systems where the brush rotation can create an uncontrolled dispersion of the removed sand with its subsequent re-deposition on previously cleaned surfaces, the brush system of the proposed robot, suitably combined with the motion and cleaning strategies, is able to control the sand remove so avoiding any possible re-deposition; (iii) the design of the brushes allows the belt-based tracks to move always on a cleaned PVP surface; (iv) unlike the other mobile systems equipped with wheels, the belt-based tracks is not affected by disturbances of the gaps between two consecutive PVPs and the PVP array mounting support; (v) in addition to the content of point (iii), the high value of the belt friction coefficient avoids the slipping of the robot; (vi) the total height of the proposed robot is lower than that one of the other reported mobile robots. This specification contributes to increase the overturning stability and to work in safety even on higher slopes of the PVP array; (vii) the real-time control of the robot position respect to a reference system fixed around the PVP array, is the design key for the autonomous drive of the robot. As a consequence, due to the regularity of the area to be cleaned and to the proposed motion strategy, the robot does not require a path recognition obtained by a 3D scanning, (see for example, the automation of earth-movers [25] or the mobile robot navigation [26]), a series of encoders and a set of smart sensors for the edge detection [19].

Another important advantage of the proposed solution is the use of a simple frame for the device fabrication and of commercially available components to implement a robot at very low cost. Moreover, the low number of the employed sensors and motors contributes to reduce even more the cost of the robot and of its maintenance. On the other hand, the proposed robot motion strategy allows for a low electrical power consumption (i.e., a low energetic robot working cost). Although the installation of external vertical barriers increases the cost of the plant, the installation and maintenance expenses of the overall proposed system are lower than those ones of the rail-based plants.

In any case, the mechanized cleaning procedures that employ brushes with soft bristles to avoid any scratches on the PVP surface, unify the advantage of manual cleaning (i.e., the reinstating of PVP to clean conditions is very effective) and, at the same time, highly reduce the labor cost.

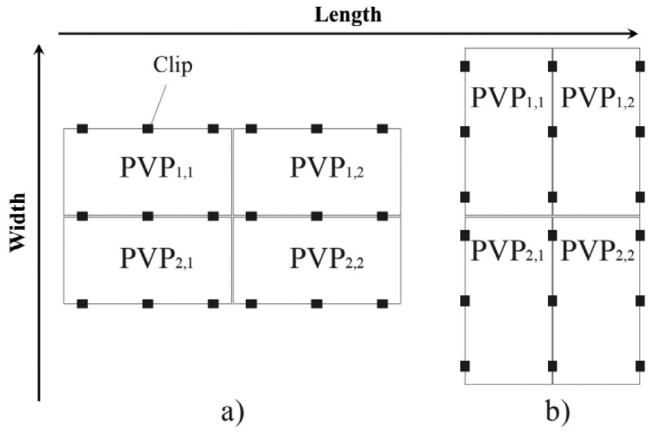


Fig. 1. Examples of a typical 2 × 2 PVP array according to a: a) horizontal and b) vertical layout.

In the following, we will discuss in detail the robot mechanical design and its motion, the cleaning strategy, the electronic control system, as the result of the mechatronic approach for a specific application [27]. Finally, the results of the experimental tests of the robot working performances in a laboratory environment will be described and discussed.

2. Background

Solar farms are constituted of sets of PVP forming a rectangular array placed according to a horizontal or vertical layout, as shown in Fig. 1. In general, there is not a standard size of the PVP since it depends on the manufacturer. In solar farms as the Desert Sunlight Solar Farm (California) or Tengger Desert Solar Park (China), four, six or eight rows of PVP are installed on frames whose length can reach up to 100 m. The PVP array width can reach 3.5 m (for the vertical layout) or 3 m (for the horizontal layout). In desert zones, the PVP array slope respect to the ground ranges from almost horizontal up to 30°, according to the solar farm longitude and latitude. Aluminum and rubber clips, whose thickness exceeds up to 6 mm the PVP surface, uniformly fix the PVP to the mounting frame. Each PVP is spaced of about 10 mm from the neighboring ones.

In desert zones, the PVP temperatures range from −10° to 70°C and, in general, with the exception of sand, no other soiling covers the PVP surface, like bird droppings and sticky stuff [11,14]. In particular, the absence of rainfalls and the reduced size of the sand particles do not create encrustations and the roughness of the PVP surfaces is those ones of commercial glasses. On the other hand, humidity [28] and dew cycles [29] exist even in the driest deserts and these ones can create a cementitious mixture on the PVP surfaces if not continuously cleaned. In any case, in the remote possibility of the occurrence of bird droppings and sticky stuff, the PVP cleaning will require a human intervention with water [30]. For these reasons, the PVP cleaning activities must be practiced continuously.

All these characteristics and constraints define the following robot features:

- the traction components (i.e. wheels, belts, tracks) must be made of a suitable high friction material to avoid sliding and slipping of the robot on the PVP surface;
- the traction system must guarantee the uphill motion up to a maximum slope equal to 30°;
- the robot must avoid the overturning up to the maximum slope;
- the sand removal must be carried out by brushes with very soft bristles to avoid scratching on the PVP surface;
- in a single rotation of the brush, the number of brushes shifts on the same areas must be sufficiently high to avoid the appearance of possible incrustations;
- the contact pressure between the robot and each PVP must be distributed on a suitable area to avoid an excessive bending of the panel that can origin uneven PVP cleaning. For this, the mass of the robot must be contained to reduce also the total power consumption;
- the seals must be conceived to avoid sand contamination of the robot moving components;
- the robot requires an on-board power supply for the self-powered traction and an embedded control system with no wirings with electrical components external to the robot;
- a parking bay must be provided for the robot recovery and for its power supply recharge.

3. The robot motion and the cleaning strategy

The definition of the robot motion and the cleaning strategy depends on two main factors: the request of a complete sand removal that avoids to wipe away the sand towards previously cleaned PVP; the reduction of the robot overall electric power consumption for all the duration of the PVP cleaning procedure. Based on these considerations, the robot has been designed to have the front and rear sections undistinguishable for what concerns the motion direction and the cleaning process. Both the robot front and rear ends terminate with a helical brush to allow the robot to move forward and backward without the need to make a rotation of 180° of the device. As a consequence, the envelope of each brush helix has been designed with opposite helix angles to allow wiping thick layers [12,14] of sand away toward PVP array areas that will be cleaned later during the cleaning process. Moreover, the brushes must alternatively work in such a way that only the brush facing along the robot motion direction will operate to wipe the sand away forwards and downwards until it falls out from the PVP surface, far from the previous cleaned PVPs surfaces. In any case, the brush length must be chosen larger than the track of the traction system to avoid the robot moving on the sand.

Fig. 2 shows the scheme of the robot motion and the cleaning procedure. During the rest periods, mainly in the daylight hours, the robot is in a sun protected isolated parking bay placed at one of the top corners of the PVP array. Apart from emergency situations (i.e. for example after a sandstorm), the PVP cleaning activity starts during the sunrise or sunset periods when the solar farm energy production is at its minimum. Even if the automatic start of the robot will not be discussed in this paper as well as the design of the parking bay, for completeness, protected cameras placed in several points of the plant and wind sensors mounted in the parking bay can be used to detect the amount of dust on the PVP array surface. Thus, this monitoring system will provide for the more efficient conditions for the starting of the robot and the beginning of the PVP cleaning procedure. Moreover, light sensors combined with a clock could be used to detect the sunset period. All the information generated by such devices can be used to give the start command for the cleaning process.

In order to explain the PVP cleaning strategy, referring to Fig. 2, we have indicated: (i) the brush operating and at rest in black and grey colors, respectively; (ii) with the continuous black arrow the current PVP cleaning paths while with the dashed black arrows the previously cleaned ones; (iii) the lane crossed by the traction system in between the dashed red lines; (iv) the orange arrows, the current direction of the wiped away sand; (v) the orange and white colored zones, the sand covered and the cleaned PVP surfaces, respectively. For sake of clarity, the parking bay appears only at the robot rest position and when it comes back after the ending of the cleaning procedure.

Remembering that during the robot motion only the brush facing moving direction is always activated, at the beginning of the cleaning procedure the robot goes out from the parking bay to perform the following actions (see Fig. 2):

Step1: the robot activates the front brush and moves horizontally, in the forward direction, along the top row of the PVP array from

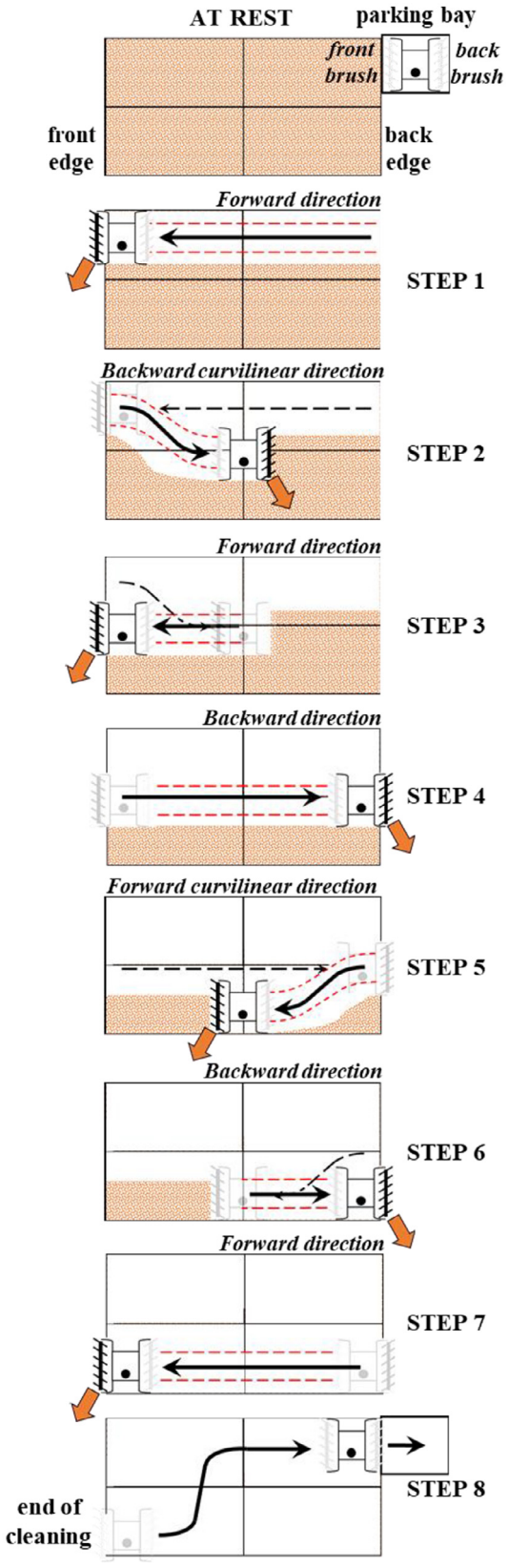


Fig. 2. The motion and cleaning strategy applied to a 2 × 2 PVP array. The orange areas are the sand covered PVP surfaces and the white areas the cleaned ones.

right to left towards the opposite corner up to reach the PVP array front edge;

Step2: at the end of the first row, the robot moves backwards to go on the next neighboring dirty row following a curvilinear trajectory and activates the back brush. At the end of this operation, the robot is horizontally aligned along the next PVP adjacent row;

Step3: the robot activates the front brush and moves toward the left direction removing the sand up to reach the PVP array front edge where it stops;

Step4: the robot activates the back brush and moves horizontally, in the backward direction, from left to right (i.e., in the opposite direction with respect to step1) stopping at the PVP array back edge;

Step5: the robot activates the front brush and moves forwards along a curvilinear trajectory going to the next PVP array row (i.e., the same action done in step2 but in a reversed motion direction);

Step6: the robot activates the back brush and moves horizontally, in the backward direction, removing the sand up to reach the PVP array back edge where it stops (i.e., the same action done in step3 but in a reversed motion direction);

Step7: the robot activates the front brush and moves horizontally along the lower row of the PVP array from right to left towards the opposite corner up to reach the PVP array front edge.

Depending on the size of the PVP array, the robot moves cleaning the PVP surfaces repeating the movements previously described in the steps from 1 to 7. When the last PVP array row is cleaned (see the "end of cleaning" panel in Fig. 2), the robot returns to the parking bay following the shortest path, according the following action:

Step8: the robot deactivates both brushes and reaches the first row (as shown in Fig. 2). At this point, the robot moves aligning itself parallel to this row, enters the parking bay and stops.

4. The mechanical design and the prototyping

The robot is moved by a couple of two independent closed belts that act like two tracks. This solution allows to: (i) better distribute the robot mass on the solar panels; (ii) enhance the traction capability; (iii) minimize the possibility of sliding and slipping; (iv) reduce unwanted motion direction changes that may occur during the PVP array row crossings or when the robot encounters the PVP array fixing clips. The mechanical design required to address the following technical issues: (i) to choose material and size of the belts; (ii) to verify the conditions for avoiding the robot overturning; (iii) to size the electric motors for the robot movement and the brush rotation. The schemes used for the mechanical design are reported in Fig. 3.

4.1. Belt design

The design of the belt in terms of material and dimensions, must consider to achieve, at the same time, the required friction of the robot with the PVP array surface and the minimum possible contact pressure exerted by the robot on the solar panel surface. A suitable friction is, in fact, necessary to avoid the robot sliding even under severe operating conditions while a restrained contact pressure preserves the PVP integrity and minimizes the bending of the solar panel surface. Moreover, the belts must be made of a material able to assure the same performances within the range of the external operating temperatures.

According to the condition of the dynamic equilibrium, the minimum static friction coefficient required f_{sMIN} depends on the slope α of the PVP array and it is equal to:

$$f_{sMIN} = \tan \alpha \tag{1}$$

The value of f_{sMIN} is equal to 0.577 for a maximum PVP array slope of α=30°. As a consequence, we have chosen two HTD® 8M closed belts made in polyurethane, whose working temperatures range from −10°C

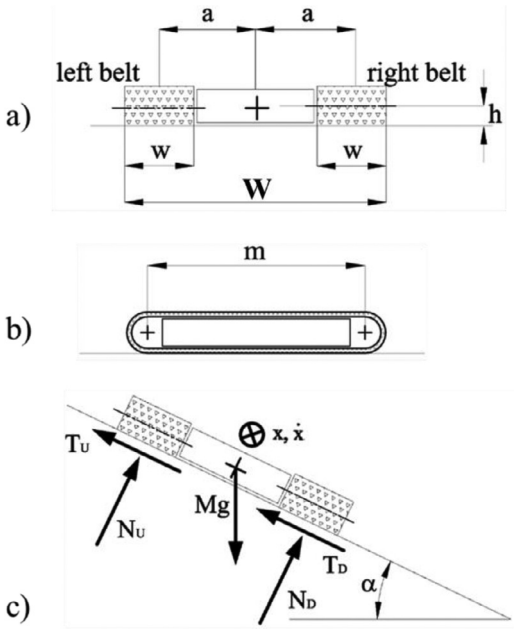


Fig. 3. Schemes used for the robot design: a) the frontal and b) lateral views on a horizontal plane with the main dimensions; c) in a plane perpendicular to both the ground and the direction of motion, the free body diagram during the horizontal motion of the robot on the α inclined PVP array.

to +80°C with peaks up to 110°C [31]. The belts are externally covered by a 2.4 mm thick layer of Linatex®. Polyurethane useful to reduce the friction between the belt and the pulley assuring the same performances in the operating temperature range. Linatex® is a high friction red colored natural rubber with friction coefficient ranging from 1 to 1.65, depending on the contact material. Moreover, Linatex® can work in the required temperature range because it is resistant up to temperatures equal to 70°C with an excellent wear and abrasion resistance [32].

On the other hand, referring to Fig. 3a and b and considering a robot overestimated mass equal to 20 kg, the belt width w equal to 84 mm and the inter-pulley length m greater than 216 mm provide for a convenient contact pressure equal to about one fifth of the maximum supported load of 5400 Pa, as reported in the datasheet of the used PVPs.

4.2. Overturning condition

The assessment of the overturning conditions was carried out according to the dynamic characteristics of the robot, placed on a PVP array inclined of an angle α with respect to the horizontal plane. During the cleaning procedure, the robot mainly moves along a straight line at a constant cruise speed whose maximum value is equal to about 0.3 m/s. Acceleration and deceleration linear ramps, whose maximum values are equal to 0.1 m/s^2, act during the robot starting and stopping phases. The curvilinear trajectories (see Fig. 2 and the description in Section 3) are traveled at lower speed so that the centrifugal and tangential accelerations can be neglected. Referring to Fig. 3c, the equations for the dynamic equilibrium are:

$$N_U + N_D = Mg \cdot \cos\alpha \tag{2}$$

$$T = T_U + T_D = Mg \cdot \sin\alpha \tag{3}$$

$$N_U \cdot a - N_D \cdot a + T \cdot h = 0 \tag{4}$$

where N_U and N_D are the resulting Up and Down normal reaction forces between the solar panels and the robot belts, T_U and T_D the Up and Down friction forces for each belt and M and g the robot mass and the gravity acceleration, respectively. Moreover, as shown if Fig. 3a, h is the half of the robot height and a the distance between the central axes of the robot and the center of each pulley. The size of a and h must satisfy the overturning condition for a defined value of the PVP array slope angle. By arranging the previous equations, we obtain:

$$N_U = 1/2 \cdot Mg \cdot \left[\cos\alpha - (h/a) \cdot \sin\alpha\right] \tag{5}$$

$$N_D = 1/2 \cdot Mg \cdot \left[\cos\alpha + (h/a) \cdot \sin\alpha\right] \tag{6}$$

While the value of N_D is always positive, the condition $N_U \geq 0$ must be verified to avoid the robot overturning, so the threshold value is $N_U = 0$. To check the value a/h for the PVP array with the (maximum) slope angle fixed at α=30°, the following condition must be verified:

$$a/h \geq \tan\alpha = 0.577 \tag{7}$$

Based on this equation and on the robot operative duties, the height of the prototype equal to $2h$=120 mm and the dimension a=200 mm were positive checked: the ratio a/h=3.33 corresponding to the overturning condition for a PVP array slope equal to about α=73° is, in fact, much higher than 30°.

The resulting distance between the lateral end of the robot belts, W of Fig. 3a, is equal to 484 mm.

Table 1
Input data for the calculus of P and C_0.

Symbol	Parameter	Value	Unit
L	length of the belt	0.610	m
f	fictive friction coefficient	0.1	
g	gravity acceleration	9.81	[m/s^2]
M_R	overall mass of the pulleys	5.51	[kg]
M'_G	belt mass	3	[kg/m]
M'_L	maximum load moved	24.60	[kg/m]
M	overall Mass of the robot	15	[kg]
α	slope of the PVP array	30	[°]
C	correction value (function of L)	3	
v	speed of the belt	0.3	[m/s]
η	efficiency of the motor	0.9	
R	radius of the pulley	0.05	[m]
P	overall power	36	[W]
C_0	overall torque	3	[Nm]

4.3. Electric motors dimensioning

The dimensioning of the two electric motors was carried out by using the model of a roller track that is able to move the robot overall mass along a slope equal to α. According to the Standard DIN 22101 "Continuous conveyors – Belt conveyors for loose bulk materials – Basis for calculation and dimensioning", the forces occurring at the belt-track system are:

$$F_H = L \cdot f \cdot g \cdot \left(M_R/L + 2\left(M'_G + M'_L\right) \cdot \cos\alpha\right) \tag{8}$$

$$F_{St} = Mg \cdot \sin\alpha \tag{9}$$

$$F_N = (C-1) \cdot F_H \tag{10}$$

where F_H is the primary resistance of the upper and lower parts of the belt due to the friction of the bearings, F_{St} the slope resistance of the conveyed load and F_N the secondary resistance due to the belt bending resistance. For the definition of the other parameters inserted in the Eqs. (8)–(10), refer to Table 1. The needed overall power P and torque C_0 can be calculated with the following expressions:

$$P = \left(F_H + F_{St} + F_N\right) \cdot v/\eta \tag{11}$$

$$C_0 = P \cdot R/v \tag{12}$$

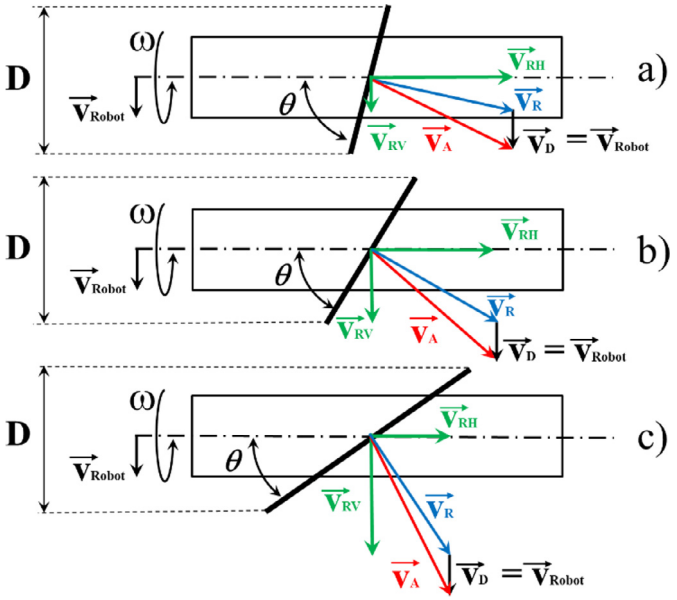


Fig. 4. Effect of the angle of the helix on the velocity of the sand. The dimensions of the vectors are not representative of the real velocity.

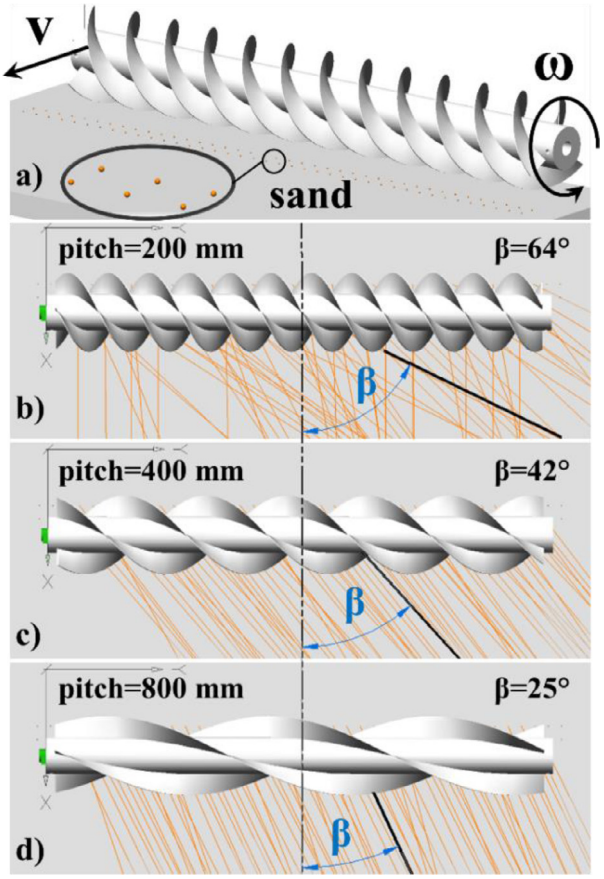


Fig. 5. The numerical model of the brush design: a) the initial condition and in the inset the placement of the simulated sand particles; the trajectories of the sand particles for different values of the brush pitch: b) 200 mm; c) 400 mm; d) 800 mm.

Assuming the values of the robot constructive parameters reported in Table 1, the values P=36 W and C_0=3 Nm have been obtained. The results overestimate the electrical motor characteristics because they refer to: (i) only one motor needed to move the overall structure of the robot; (ii) a prototype whose frame and pulleys are made of aluminum, heavier than a polymeric material whose robot must be made of. According to these results, the robot uses two OSLV MR 62 DC electric gear-motors having each one a maximum power of 39 W, a torque of 2 Nm at the operating voltage of 12 VDC and a nominal speed of 189 rpm.

4.4. Brushes dimensioning

The length of each brush was fixed equal to 800 mm, 1.65 times the dimension of W (484 mm) as shown in Fig. 3a. This choice avoids the belt-based track-wheels moves on the sand: during the robot rotation (see Fig. 2), the brushes create a lane wider than that one crossed by the belts. The number of passages of the robot on the PVP needed to completely clean its surface, depends on the dimensions of the brush width. On the basis of the used PVPs (dimensions 1628 × 844 mm), the robot must pass on the same PVP at least two times, for both horizontal and vertical layouts. To enhance the effectiveness of the sand removal, the brush has been conceived to be equipped with three helixes of bristles angularly spaced at 120°.

To design the brush, some preliminary aspects have been considered from a kinematic point of view. To wipe away the sand, the peripheral brush speed $v_P = \omega \cdot D/2$ (with ω and D the angular speed and the external brush diameter, respectively) must be much higher than the linear robot speed v_{Robot}. Then, referring to Fig. 4, the absolute sand velocity v_A depends mainly on the relative velocity v_R (with respect to the brush), whose value depends on ω. Finally, the movement of the sand at one side depends on the horizontal component of the relative velocity v_{RH}. Such component increases as θ increases. For high values of θ, the sand is predominantly moved at one side and slightly forward. Over a certain value of θ, the sand could be moved between the helixes with a consequent accumulation within the brush. Low values of θ, corresponding to an increase of the vertical component of the relative speed v_{RV}, produces a predominant forward movement of the sand and slightly at one side. Under a certain value of θ, considering the contribution of the dragging speed v_D, the sand could be moved only in the forward direction. This last aspect must be avoided because it would cause an accumulation of the sand on the next PVP surface, could wipe away sand on the previously cleaned PVPs and would not take profit from the PVP array slope.

The brush diameter D has been set equal to 130 mm. Considering the relationship among D, θ and the pitch, instead of θ only the pitch has been considered for the brush design. Then, in order to find the proper value of the pitch, a numerical simulation model has been developed with the aim to analyze the trajectory of the sand after the contact with the rotating brush. Such trajectory depends on the value of the pitch, the angular brush speed ω and the robot speed v_{Robot}. The proper value of the pitch is that one that guarantees to obtain the wiped away sand trajectory (i.e. in the forward direction and at one side).

The numerical model has been implemented by the software SimWise 4D used for the study of the kineto-dynamics of multi-body systems. A rigid body model has been employed because of the high bending stiffness of the chosen bristles of the real brush. As shown in Fig. 5a, a single layer of sand has been modeled composed of a set of spherical rigid particles with a diameter equal to 1 mm and mass of 0.1 g. This single layer has been placed on a rigid ground along two parallel lines in front of the brush. The latter has been modeled by a rigid cylindrical body wrapped by three rigid continuous helixes angularly spaced at 120°. The linear speed v, equivalent to v_{Robot}, and the angular speed ω are applied to the brush that moves on the rigid ground.

Suitable contact properties have been defined between each sphere with the ground and with the brush. In order to consider the property of the sand to create agglomerates, the friction coefficients between the sand and the ground and between the sand and the brush have been fixed equal to 1.0. No slope has been assigned for the ground and the

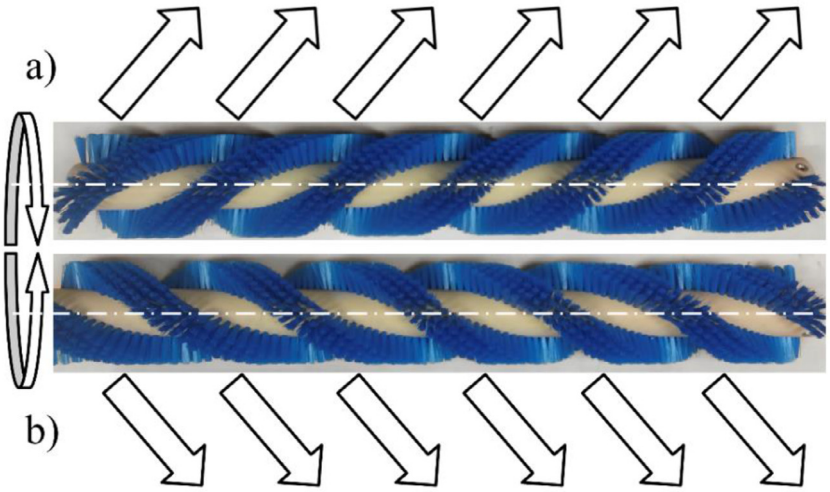


Fig. 6. The brushes.

brush because the gravity acting on the particles of sand can be neglected. Considering the robot maximum cruise speed of 0.3 m/s, as reported in Table 1, the value of v has been varied in the range 0.01 m/s (to simulate the most severe condition when the robot moves very slowly) up to 0.3 m/s. Considering the real working brush angular speed, the value of ω has been set equal to 850 rpm.

Several simulations have been carried out with models of brushes with pitch values varying from 200 mm to 800 mm with a step of 50 mm. Fig. 5b–d show only three configurations at v equal to 0.01 m/s. It can be noticed that the increase of the pitch reduces the angle β of the sand particles trajectory with respect to the robot motion direction. As a consequence, high values of the pitch could provide for a predominant forward motion of the sand (Fig. 5d). For the pitch equal to 200 mm, the angle β reaches the maximum value (Fig. 5b). Nevertheless, the sand particles do not follow the same trajectory: some of them are directly wiped away, out the working volume of the brush; some others are wiped away, within the brush and then out of its working volume of the brush. This pitch value causes unpredictable sand trajectories. Good results have been achieved for a pitch equal to 400 mm (Fig. 5c): the sand particles follow almost the same trajectory, sufficiently angled respect to the robot motion direction and safe for what it concerns the direct sand wiping away. For values of v higher than 0.1 m/s, the simulations show that the sand is directly wiped away just starting from the lowest chosen value of the pitch. Based on the results of the simulations, the achieved optimal value of the pitch has been set equal to 400 mm corresponding to θ equal to 45.6°.

From the constructive point of view, the single helix is made of four tufts lines of 0.35 mm diameter bristles made of polybutylene terephthalate chosen to improve the effectiveness of the sand removal, to avoid the scratching of the PVP surface and to resist to many cleaning procedures. The resulting mass of the brush is lower than 2 kg and its rotation is accomplished by a Como Drill 975D41 DC electric gear-motor with a maximum power of 41.3 W, torque C_m of 0.28 Nm at the DC voltage of 12 V and nominal speed of 1750 rpm. A T10 polyurethane belt transmission between the motor and the brush has been used with a speed ratio equal to 1. A direct coupling motor-brush has been excluded to contain the size of the robot and to reduce the bending effect of the motor on the supporting structure of the brush.

The brushes adopted for the robot and their operating conditions are shown in Fig. 6 where the rounding arrows indicate the rotation directions of the brushes. With reference to Fig. 2, the front brush work when the robot moves from left to right (a) and from right to left (b). The straight arrows represent the direction along which the sand is wiped away from the PVP surface.

4.5. Robot prototype

The robot prototype has been constructed by employing 3 mm thick aluminum sheets for the effortless availability of commercial components such as profiles and pulleys and for the simplicity of machining. At the industrial and production level, the robot will be produced with the structure and the mechanical components made by polymeric materials. As reported in Table 1, the actual robot prototype mass is equal to 15 kg that is more than 3 times the estimated mass of the robot fabricated with polymeric material. Fig. 7a reports the views of the structure of the robot prototype with the main dimensions in millimeters. As shown in Fig. 7, the robot is the result of the assembly of a central unit, two traction elements and two cleaning modules. The central unit collects the battery for the power supply of the on-board electronic control system, the high-resolution ultrasound sensors network for the autonomous motion of the robot, the motors of the traction elements. The central unit is a box open on the top side whose dimensions are 400 mm (width), 630 mm (length) and 104 mm (height). The two tracks are equal and symmetrically mounted along the longer sides of the central unit and fixed by a set of screws. Each of the traction elements is made by using a C-shaped aluminum profile (100 × 100 mm with a thickness of 4 mm) with the open side placed at the bottom of the prototype, as shown in Fig. 7b. One driving and one idle pulley equipped with ball bearings suitable fixed on the vertical edges of the C-shaped profile, are located inside the traction element. Around these two pulleys is wrapped the transmission belts. The overall length of each belt is equal to 1200 mm and the distance m measures 440 mm. The driving pulleys, one for each track, are directly connected to the electric motor for enabling the torque transmission. The motor fixing systems are mounted inside the central unit on its lateral sides. The bottom edges of the resulting assembly of the central unit facing the PVP array surface were covered by nylon strip-brushes mounted using screws in order to avoid sand contamination of the moving parts of all the traction components. The two cleaning modules are equal and symmetrically mounted by screws at each end of the traction modules. Each brush is fixed to a shaft whose supports, Y-bearing oval flanged units, are mounted on a couple of L-shaped cantilevers made by commercial aluminum profiles. On one of the cantilever are placed the gear-motor and the transmission system to perform the brush rotation. Two aluminum plates with a thickness of 0.5 mm close the cantilevers and a silicon rubber film was used to seal the cleaning modules to avoid the sand contamination of the transmission moving parts. Moreover, the cantilevers are equipped with slots to adjust the proper height of the brushes respect to the PVP array surface. The realized prototype is shown in Fig. 7c.

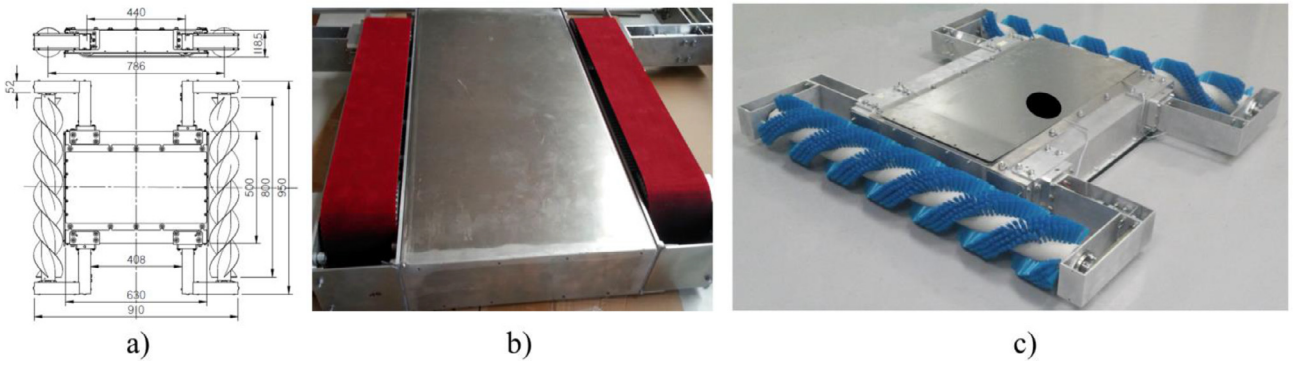


Fig. 7. The robot prototype: a) the lateral and top views with the sizes in mm; b) bottom view; c) overall top view (black circle to be compared to Fig. 2).

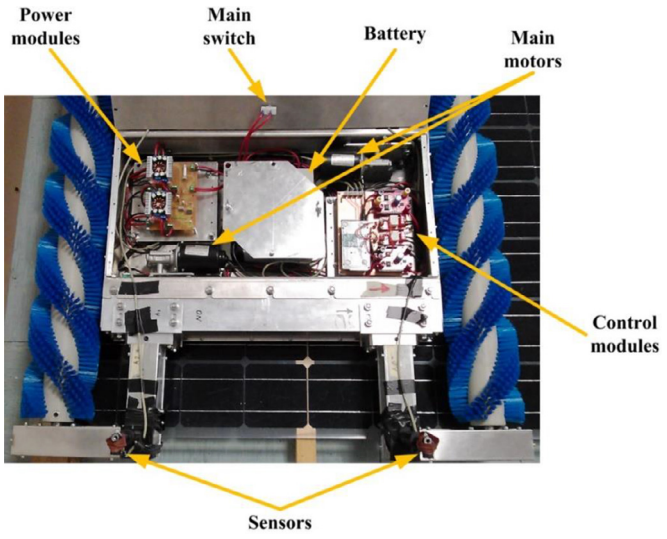


Fig. 8. The robot overall electronic control system together with the main motors and the battery.

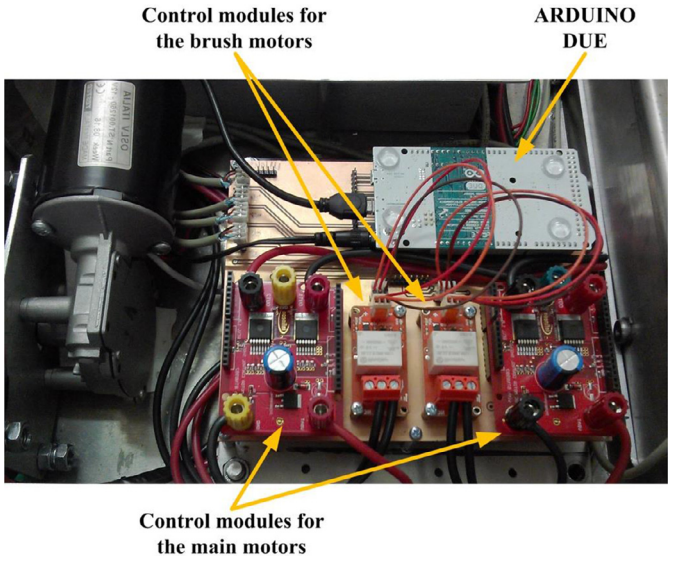


Fig. 9. The PCB, the ARDUINO DUE and the shields for controlling the motors and processing the sensor electronic signals.

5. The electronic control and powering system

As shown in Fig. 8, the overall electronic circuitry is mounted inside the robot, together with the two main motors (for the robot motion) and the battery delivering the power supply. The developed electronic circuitries have been designed, simulated and prototyped (i.e., PCB layout) in the OrCAD PSpice environment. The electronic circuitries allow to autonomously move the robot, regulate its position, direction and speed rate, properly activate and regulate the brush rotation speed, manage the power supply of the robot and, finally, control the battery charge level.

In particular, the on-board control system employs an ARDUINO DUE board based on a 32-bit ARM core microcontroller (i.e., Atmel SAM3 × 8E ARM Cortex-M3 CPU). It is mounted on a PCB that includes also the modules to manage and regulate the motors and to process the electrical signal coming from the sensors as reported in Fig. 9.

The robot is powered through a 12V DC, 22Ah REC22-12I valve-regulated lead acid battery by YUASA. In this regard, a second PCB has been developed for the supply voltage management of the complete circuitry. The implemented schematic circuit is reported Fig. 10 and the related fabricated PCB is shown in Fig. 11. The circuit includes: 1) the main power switch; 2) an input protection 6 × 32 mm 15 A 250 V slow-blow ceramic fuse; 3) two power control modules 15 A DC/DC converter by DROK (input 4-32 V, output 1.2-32 V) as voltage regulators providing the stable supply voltages to the electronic circuits and, in particular, to the two motors (11.5 V and 6 V for the robot motion and brush rotation motors, respectively); 4) a battery charge monitoring/control unit with voltage level indicator and protection circuit that automatically communicates to the control unit to drive the robot towards the PVP array parking bay in the case of high current consumptions and/or low battery level.

The ARDUINO-based platform has been suitably programmed to manage all the functions that the robot must perform such as the moving and cleaning operations and its reposition in the parking bay. More in detail, as previously described, the robot motion is achieved through two electric gear-motors (OSLV MR 62 DC) whose speed and direction of rotation are regulated by means of two external motor control shields based on high current PN half-bridge NovalithIC BTN8982TA by Infineon. These two motor drivers, controlled through PWM-based signals provided by ARDUINO DUE board, act on the two DC gear-motors in a bi-directional mode (i.e., in the forward and backward directions). The other two DC gear-motors (Como Drill 975D41) used for the rotation of the cleaning brushes are simply activated/deactivated by means of two 250 V 10 A electro-mechanical relays modules T010010 by TinkerKit through 0-1 (i.e., on-off) digital signals, always generated by the ARDUINO DUE board. Moreover, four high-resolution ultrasonic range finders (MB1023 HRLV MaxSonar EZ Series) have been used for the real-time detection of the current position of the robot as well as to adjust its speed and motion direction. These sensing devices, powered at +5V DC single supply, are connected to the analogue inputs of ARDUINO DUE board and mounted on the four corners of the robot, on the upper

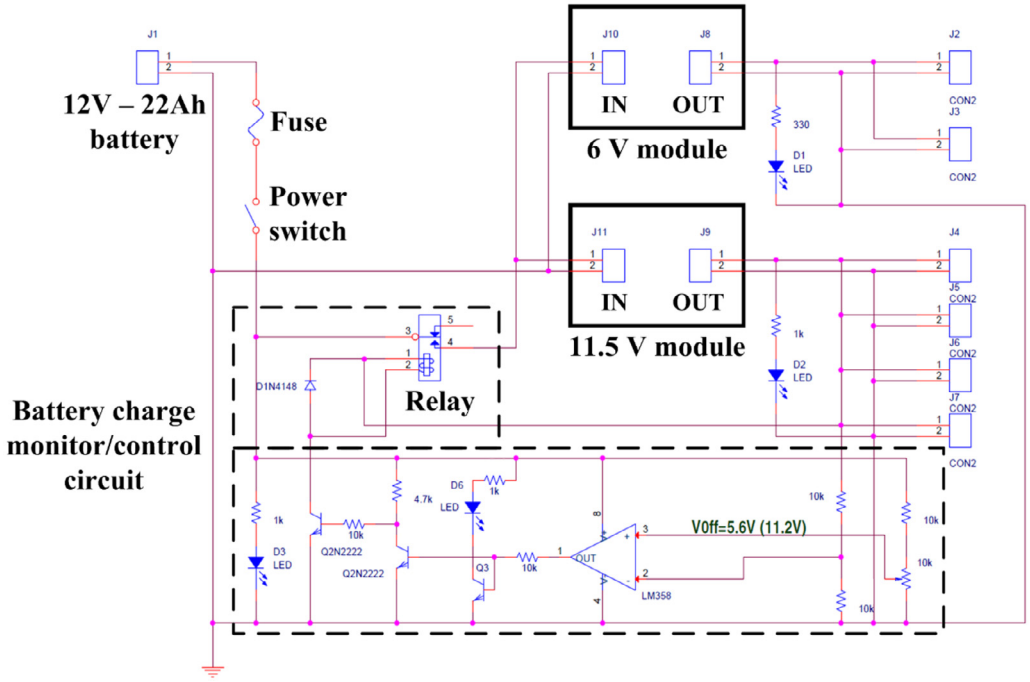


Fig. 10. Schematic of the robot power supply management circuit.

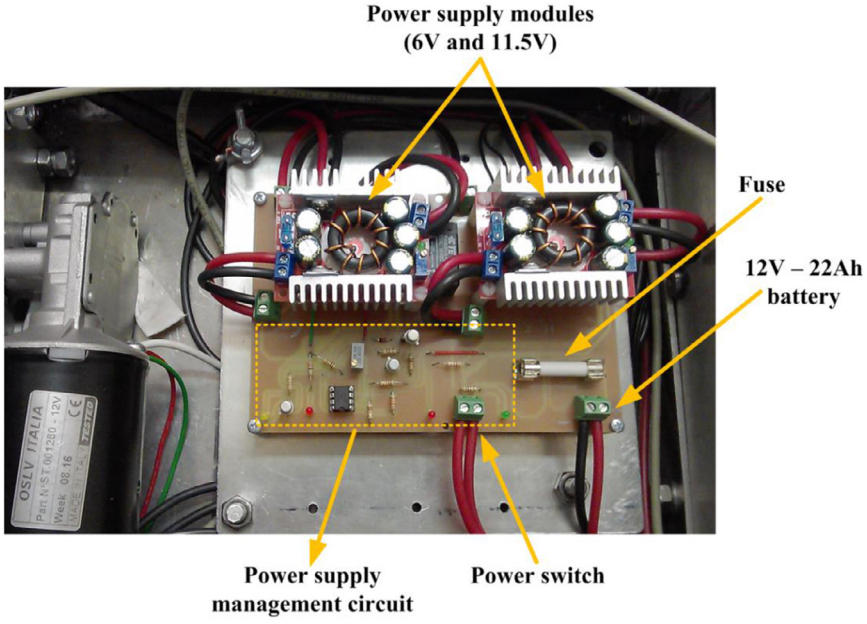


Fig. 11. The PCB for the management of the robot power supply.

surface of the L-shaped cantilevers of the cleaning modules, to detect all the needed spatial parameters (i.e., front, back, left and right distances from appropriate targets) to evaluate the actual robot position. The sensors provide analogue output voltage signals proportional to the measured distances between the robot and the supplementary vertical reference barriers externally mounted around the PVP array perimeter, to allow the passage of the robot without the occurrence of accidental contacts. In the present study, the height of the vertical barriers has been set equal to 200 mm, since the ultrasonic sensors have been vertically mounted for an easier calibration. Nevertheless, the engineered robot will include the ultrasonic sensors inside its structure so reducing the height of the vertical barriers to about 20-30 mm and, consequently, avoiding their possible shadows on the PVPs array surface. This will reduce also the impact of the wind and the installation costs. The used ultrasonic sensors have internally a speed-of-sound temperature compensation that allow for a high accuracy proximity detection from 30 cm to 5 m with 1 mm resolution [33]. By means of all these devices, the complete ARDUINO-based platform allows controlling the overall timing of the any robot actions and motions, starting from its initial position located in the parking bay.

6. The algorithm for the robot motion control and the cleaning procedure

In Fig. 12 we report the schematic of the PVP array used for the laboratory experimental verifications of the robot motion and cleaning procedure. The array is composed of 2×4 PVPs and at its upright corner

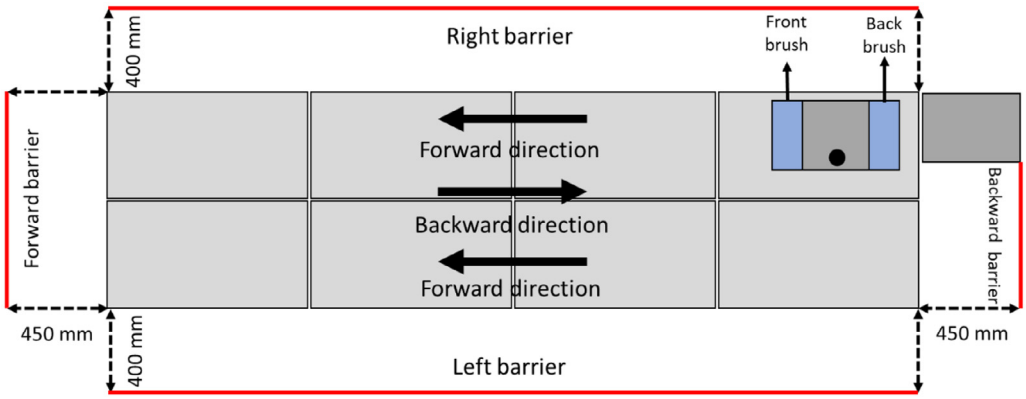


Fig. 12. Schematic of the PVP array used for the laboratory experimental verifications of the robot motion and PVP array cleaning procedure (black circle as in Fig. 2).

is located the parking bay that hosts the robot at the beginning and ending of the cleaning procedures. As detailed above, around the PVP array perimeter are located the vertical barriers: the left and right barriers (front and back barriers) are positioned 400 mm (450 mm) away from the PVP array edges. The distance of the front and back barriers from the PVP edges has been chosen to allow for the complete cleaning of the PVPs that is obtained by permitting the brush to overpass the PVP edge so achieving the fall down of the sand. By considering the brush length, the complete cleaning of the PVP array surface of Fig. 12 is achieved if the robot, starting from the parking bay, travels three times along the entire length of the array rows (i.e., three complete cleaning paths): two times forward (FW) and one time backward (BW). In this way, at the end of the cleaning procedure, the robot will be located at the low left corner of the PVP array. From this position, the robot must travel toward the parking bay waiting for a new start command. As reported in Fig. 2, each time the robot arrives at the end of a PVP cleaning path, it must be repositioned on the next parallel PVP cleaning path by following a curvilinear trajectory.

As stated above, all the operations needed for a correct robot motion and cleaning procedure are monitored by the ARDUINO DUE board that executes a motion control algorithm through a developed programming code based on the flowchart reported in Fig. 13. In the START (INITIAL SETTINGS), CHECK START OPERATIONS and SOFT START blocks the robot is initially in the parking bay and, once received the command to proceed to accomplish the cleaning procedure, the motion control algorithm: (i) sets the initial lateral distances from the left and right barriers; (ii) enables the rotation of the FW brush; (iii) after 3 s the robot is softly moved in the FW direction until it reaches the cruise speed of 0.30 m/s. Then, the four flowchart blocks CHECK L-R POSITION, POSITION RECOVERY, CHECK CLEANING PATH END, SOFT STOP, describe the operations done by the closed loop position control algorithm, inside of the higher level motion control algorithm, to allow for the robot motion in a straight direction along the PVP array cleaning path. These operations are the same in both the FW and BW directions of the robot motion. During the robot motion, the ARDUINO DUE acquires the signals from the four on-board ultrasonic sensors every 100 ms. In particular, from the signals generated by the sensors facing the left and right barriers the position control algorithm checks the distances from these barriers.

When the robot travels along the first PVP cleaning path, these distances are equal to the initial setting values, while for the other cleaning paths the distances are programmed in such a way to obtain the optimal cleaning of the PVP array. Distances are maintained at the reference values, within the tolerance range of +/- 5 mm (i.e. the robot is travelling along the correct path) by the position control: it operates on the motor of that belt able by a step signal (i.e., by accelerating or decelerating) to take back the current distances within the tolerance range.

At the same time, the signals from the sensors facing the FW or the BW barriers are detected: when the corresponding distances from these barriers are equal or less than 900 mm, the robot is gently arrested with the FW or BW brush stopped just outside the PVP array to allow the fall-down of the sand. Indifferently if the robot traveled along the FW or BW directions, the robot must be repositioned in the following parallel cleaning path only if the distances from the left and right barriers are different from those ones corresponding to the final PVP cleaning path (i.e., the robot did not reach yet the position at the lower left corner of the PVP array that indicates the end of the cleaning procedure). This particular operation is performed by the CHECK CLEANING PATH CHANGE block of the flowchart. If this condition is satisfied, the robot also verifies if the PVP cleaning operations are concluded through the block CHECK CLEANING END. If not, the robot must be repositioned on the next parallel PVP array cleaning path by performing the operations contained in the CLEANING PATH CHANGE OPERATIONS block of the flowchart. The motion control algorithm sets the new distances from the left and right barriers, enables the FW or the BW brush and acts on the motor belts in such a way to move the robot along a programmed curvilinear trajectory to relocate it on the new PVP cleaning path. This operation ends when the distances from the left and right barriers are equal to the new reference ones. In particular, if we refer to the first FW PVP cleaning path, the first robot reposition will be performed by acting on the BW brush and along a clockwise curvilinear trajectory in the BW direction. On the contrary, at the end of the second PVP cleaning path, the robot relocation in the third PVP cleaning path is obtained by acting on the FW brush and travelling a counterclockwise curvilinear trajectory in the FW direction. Referring to Fig. 2, when the motion along the curvilinear trajectory ends, the rotating FW (or BW) brush is arrested, the BW (or FW) brush is activated and the robot moves in the BW (or FW) direction until reaching the border of the new PVP cleaning path on the same side of the previous one. Now the robot is ready to begin the final cleaning of the new PVP cleaning path by employing the control operation described in the previously detailed SOFT START, CHECK L-R POSITION, POSITION RECOVERY, CHECK CLEANING PATH END, SOFT STOP, blocks of the flowchart. As discussed above, when the cleaning procedure is completed, the robot stops at the left low corner of the PVP array and from the CHECK CLEANING PATH CHANGE and CHECK CLEANING END blocks of the flowchart the motion control algorithm executes the RETURN TO THE BAY OPERATIONS block. At this point, the motion control algorithm sets the initial setting values from the left and right barriers, moves the robot along a counterclockwise curvilinear trajectory in the BW direction until the first cleaned PVP path is reached. At this time the robot travels in the BW direction controlled again by the operations defined in the blocks SOFT START, CHECK L-R POSITION, POSITION RECOVERY, CHECK CLEANING PATH END, SOFT STOP. Once reached the parking bay, the conditions verified by the

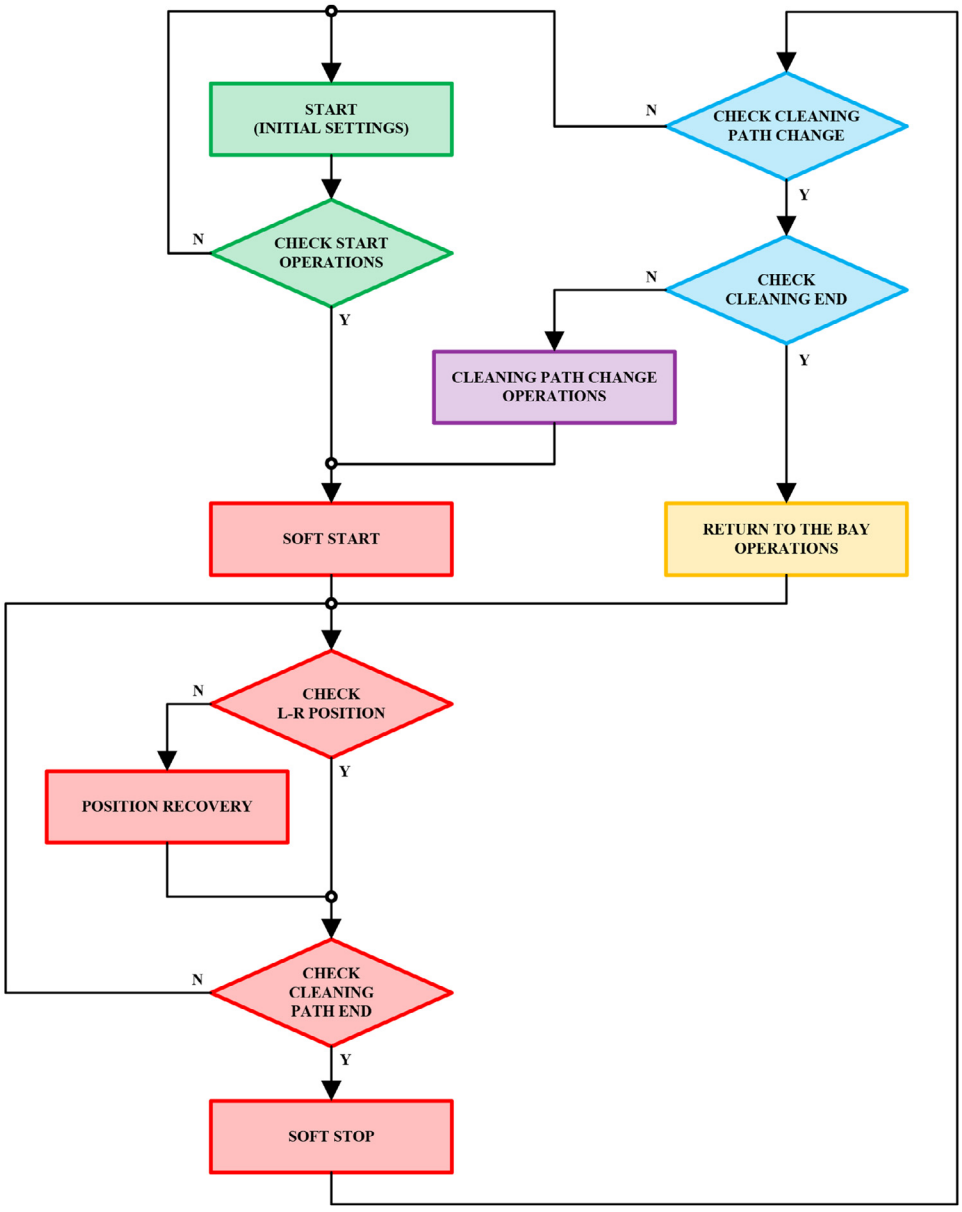


Fig. 13. Flowchart describing the motion control algorithm for the robot and the PVP array cleaning procedure.

block CHECK CLEANING PATH CHANGE are satisfied (i.e., the distances from the left and right barrier are equal to the initial setting values) the robot stops until a start command for a new cleaning procedure arrives. In the present paper, we are not interested in detailing which could be the methodology to be used to activate the cleaning procedure since it will depend on the characteristics of the solar plant and on its location.

7. Laboratory experimental tests and results

The robot prototype was subjected to laboratory experimental tests by using a 2 × 4 horizontal layout PVPs array with an overall area of about 11 m^2 considering the dimension of a single PVP panel of 1628 × 844 mm. The array was mounted on a frame equipped with two coaxial hinges for the rotation of the array allowing a PVP slope regulation in the range 0° – 30°. The maximum speed rate of the robot and of the brushes have been fixed to about 0.3 m/s and 850 rpm, respectively.

A series of laboratory tests have been carried out in order to check the absence of robot overturning at the maximum slope of the PVP array. In addition, for different slopes it was assessed also the capability of the prototype to move upwards following the operative motion direction. In these working conditions, the robot was intentionally tilted for different angles with respect to the operative motion direction (i.e., deviations of the robot from the straight cleaning direction) so to evaluate the time needed to perform the self-reposition. Fig. 14 summarizes the experimental results related to the measurements of the self-reposition time of the robot for different slopes of the PVP array (i.e., 0°, 10°, 20° and 30°) and tilt angles equal to 10°, 20° and 30° from the robot motion direction. The evaluated recovery times range from a minimum of 2.5 s to a maximum of 14.5 s. Furthermore, it has been also verified that the robot is capable to perform over than 50 consecutive complete laps corresponding to a total cleaned surface area of about 550 m^2 (i.e., 11 m^2 x 50) without any significant deviation from the correct motion direction, stopping perfectly at the end of each PVP array edge and correctly returning in the parking bay at the end of the cleaning procedure. Table 2 reports the experimental results concerning the cleaning times of the robot, at the fixed tilt angle equal to 0°, moving along a PVP array cleaning path (with total length of about 6.5 m) as a function of the PVP array slope. The measurements show that the average cruise speed decreases from 0.26 m/s to 0.21 m/s for PVP array slopes equal to 0° and 30°, respectively. These data correspond to an increase of the av-

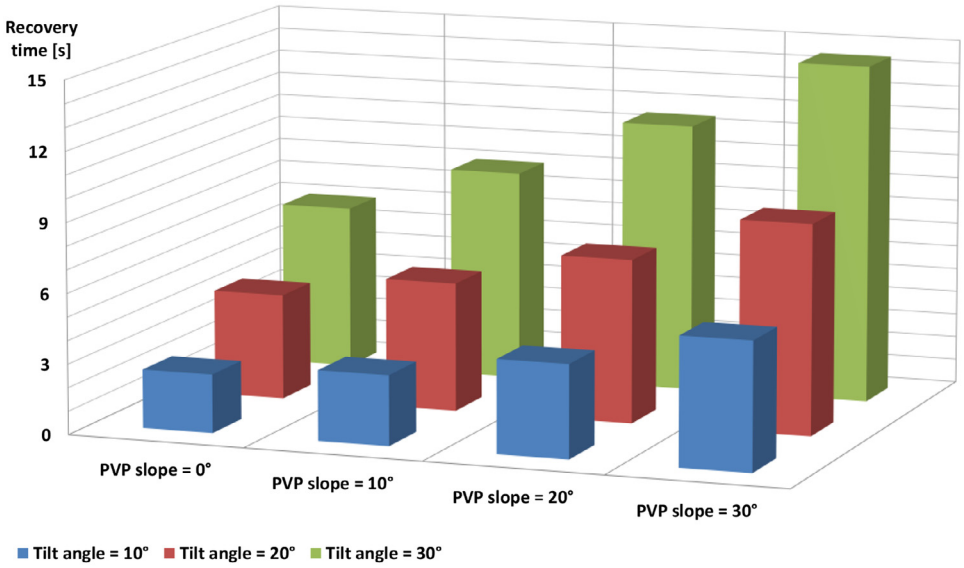


Fig. 14. Experimental laboratory tests: resulting recovery time of the correct self-reposition of the robot along the operative motion direction for tilt angles equal to 10°, 20° and 30° and for PVP array slopes equal to 0°, 10°, 20° and 30°.

Table 2
Experimental laboratory tests: resulting cleaning times of the robot moving along a PVP cleaning path (length of about 6.5 m) as a function of the PVP array slope.

PVP array slope [degree]	Average cleaning time [s]	Average cruise speed [m/s]
0	25.0	0.26
10	26.3	0.25
20	28.3	0.23
30	31.2	0.21

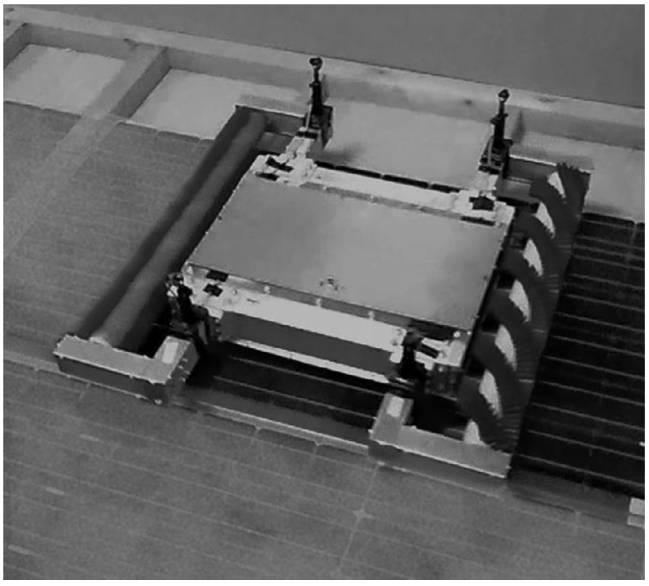


Fig. 15. The robot operating a cleaning operation of the PVP array surface.

erage cleaning time of about 30 % with respect to the robot operating conditions at a slope of 0°. Finally, laboratory cleaning tests have been scheduled after dispersing a few millimeter-thick sand layer on the PVP array. In Fig. 15 is shown a photogram of the robot during one of the cleaning operations. The sand was wiped away forwards and downwards according to the design specifications and the cleaning of the solar panel was satisfactory. Moreover, we have verified that no contamination of the moving mechanical parts with the sand occurred.

8. Conclusions

In this paper we reported on the capabilities of a self-guided robot to perform cleaning procedure of PVP arrays. In particular, the robot has been designed to autonomously operate in severe environmental conditions like those ones occurring in desert zones where very large solar plants are located. The robot is equipped with two motorized tracks for the motion. The two brushes located at the two ends of the robot along the motion direction enable the PVP array cleaning operation. An on-board electronic control system, whose input signals come from ultrasonic sensors, provides for the real time detection and the regulation of the position and speed rate of the robot.

Respect to other systems reported in Literature developed to accomplish waterless PVP array cleaning procedures, the proposed robot does not need any guide rails or pins or suction cups to move on the PVP array. The used brush geometry and its rotation conditions together with a suitable developed cleaning procedure, allow for a robot travel always on cleaned PVP surface and for a controlled sand removal that avoids any possible its re-deposition on the PVP surface. For what concerns the robot traction solution, the high value of the belt friction coefficient combined with a small total robot height, avoid the slipping of the robot up to a slope angle of the PVP array respect to the horizontal plane of 30° and largely increase the overturning stability.

Experimental laboratory tests have been carried out on a 2×4 PVP array whose slope can be adjusted in the range from 0° to 30° respect to the horizontal plane. In particular, the robot has been intentionally misaligned respect to the right straight motion direction for angles up to 30° for different slope angles. Under these conditions the measured times to self-recovery the straight directions range from a minimum of 2.5 s to a maximum of 14.5 s. Finally, it has been verified the continuous operation of the robot that was capable to perform over than 50 consecutive complete laps of the laboratory PVP array covering a total surface area of about 550 m^2 without any inconvenience.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Michele Gabrio Antonelli: Conceptualization, Methodology, Investigation, Software, Validation, Writing - original draft, Writing - review & editing. **Pierluigi Beomonte Zobel:** Conceptualization, Methodology, Investigation, Writing - original draft, Writing - review & editing. **Andrea De Marcellis:** Conceptualization, Methodology, Investigation, Software, Validation, Writing - original draft, Writing - review & editing. **Elia Palange:** Conceptualization, Methodology, Investigation, Software, Validation, Writing - original draft, Writing - review & editing.

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